P.7 SCIENCE

LESSON PLANS

TERM 1 2026

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PRIMARY SEVEN SCIENCE LESSON PLANS TERM ONE 2026

LESSON PLAN 1

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Definition of key terms: cell, tissue, organ, system, skeletal system, skeleton

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COMPETENCES:

- **Subject Competence: The learner defines key biological terms related to the muscular-skeletal system and distinguishes between them.**
- **Language Competence: The learner uses scientific vocabulary correctly in sentences (e.g., "A tissue is a group of cells performing a similar function.")**

TEACHING METHODS/TECHNIQUES:

- **Question and answer**
- **Demonstration**
- **Guided discovery**
- **Use of charts**
- **Discussion**
- **Note-taking**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Define a cell, tissue, organ, and system**
2. **Define skeletal system and skeleton**
3. **Arrange the levels of organization in order**

TEACHING/LEARNING MATERIALS:

- **Chart showing levels of organization (cell → tissue → organ → system)**
- **Chart of human skeleton**
- **Chalkboard/whiteboard**
- **Learners' books**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners warmly - Shows a brick and asks: "What is a building made of?" - Relates bricks to cells and building to body - Asks: "What do you think is the smallest unit of a living thing?"	- Question and answer - Analogy - Brainstorming	- Responds to greeting - Observes the brick - Answers questions - Suggests ideas about smallest unit
20 mins	EXPLORE	- Displays chart showing levels of organization - Asks learners to observe the chart in pairs - Asks: "What do you notice? How are these arranged?" - Lets learners discuss and share observations - Provides additional examples of each level for discussion	- Pair work - Guided discovery - Observation - Group discussion	- Observes the chart carefully - Discusses in pairs - Shares observations with class - Asks questions - Suggests additional examples
30 mins	EXPLAIN	- Defines each term clearly: - Cell: basic unit of a living thing - Tissue: group of living cells specialized to perform a similar function - Organ: group of body tissues specialized to perform a similar function - System: group of body organs specialized to perform a similar function - Skeletal system: body system that gives an animal support, shape and ability to move - Skeleton: structure that supports the body of an animal - Human skeleton: framework of bones in the human body - Writes definitions on board - Gives examples of each level - Provides diagrams for each level of organization - Explains the hierarchy from cell to organism	- Lecture method - Demonstration - Use of charts - Note-taking - Question and answer	- Listens attentively - Copies definitions - Gives own examples - Asks questions - Draws diagrams in books

15 mins	**ELABORATE**	**- Asks learners to arrange the levels from simplest to most complex - Asks: "Which level comes first? Which comes last?" - Guides learners to see the hierarchy: cell → tissue → organ → system → organism - Gives more examples of organs and systems - Asks learners to identify organs in their own bodies - Discusses how different systems work together**	**- Question and answer - Guided practice - Application - Discussion**	**- Arranges levels in order - Answers questions - Gives more examples - Writes in books - Identifies organs on own body - Participates in discussion**
5 mins	**EVALUATE**	**- Asks oral questions to check understanding: - "What is a cell?" - "What is a tissue?" - "What does the skeletal system do?" - Gives class work**	**- Oral assessment - Written assignment**	**- Answers oral questions - Copies class work**
5 mins	**EXTEND**	**- Tells learners to find out what a human skeleton looks like - Asks them to draw and label a simple diagram of the skeleton at home - Prepares them for next lesson on types of skeleton**	**- Home assignment - Reinforcement**	**- Listens to home task - Prepares for next lesson**

CLASS WORK:

1. **What is a cell?**
2. **Define a tissue.**
3. **What is an organ? Give two examples.**
4. **What is a system? Give one example.**
5. **What is the function of the skeletal system?**
6. **Arrange the following in order from simplest to most complex: organ, cell, system, tissue**

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 2

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Types of skeleton: Exoskeleton, Hydrostatic skeleton, Endoskeleton

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COMPETENCES:

- **Subject Competence: The learner identifies and describes the three types of skeletons found in animals.**
- **Language Competence: The learner names animals with each type of skeleton and explains key features.**

TEACHING METHODS/TECHNIQUES:

- **Use of charts/pictures**
- **Demonstration**
- **Question and answer**
- **Classification**
- **Discussion**
- **Note-taking**

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. **Name the three types of skeletons**
2. **Give examples of animals with each type**
3. **Explain the term moulting and give examples of animals that moult**

TEACHING/LEARNING MATERIALS:

- **Chart showing animals with different skeletons**
- **Pictures of insects, crabs, earthworms, humans**
- **Real specimens (if available): crab shell, earthworm**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Shows a crab shell (or picture) - Asks: "What is this? Where does it come from?" - Asks: "Do humans have something like this outside?" - Introduces topic: types of skeleton	- Realia - Question and answer - Brainstorming	- Responds to greeting - Observes crab shell - Answers questions - Listens
20 mins	EXPLORE	- Displays chart showing three types of skeletons - Divides learners into groups - Asks each group to observe and list characteristics of each type - Asks: "Which animals have skeleton outside? Which have inside? Which have no hard skeleton?" - Lets groups present their findings	- Group work - Guided discovery - Observation - Presentation	- Observes chart - Discusses in groups - Lists characteristics - Presents findings to class
30 mins	EXPLAIN	- Explains each type of skeleton: - Exoskeleton: skeleton found outside the body; made of cuticle; found in arthropods (insects, crustaceans, myriapods, arachnids) - Hydrostatic skeleton: body filled with fluid that produces pressure to change shape and aid movement; found in earthworms, slugs, jellyfish - Endoskeleton: skeleton found inside the body; made of bones; found in vertebrates (humans, fish, birds, reptiles) - Explains moulting/ecdysis: shedding of cuticle to enable growth; influenced by hormone ecdysone - Lists animals that moult: houseflies, crabs, spiders, millipedes, shrimps - Draws diagrams of animals with each skeleton type	- Lecture method - Use of charts - Question and answer - Drawing - Note-taking	- Listens attentively - Takes notes - Names examples - Asks questions - Copies diagrams

15 mins	**ELABORATE**	**- Asks learners to classify given animals into the three skeleton types** **- Gives list: human, grasshopper, earthworm, crab, fish, snail, bird, spider, lobster, jellyfish** **- Guides learners to write answers in a table** **- Discusses advantages and disadvantages of each skeleton type** **- Asks: "Why do insects need to moult?"**	**- Classification** **- Guided practice** **- Application** **- Discussion**	**- Classifies animals** **- Creates classification table** **- Writes in books** **- Answers questions** **- Participates in discussion**
5 mins	**EVALUATE**	**- Asks oral questions:** **- "Name the three types of skeletons"** **- "What is moulting?"** **- "Give two animals that moult"** **- "What type of skeleton do human beings have?"** **- Gives class work**	**- Oral assessment** **- Written assignment**	**- Answers oral questions** **- Copies class work**
5 mins	**EXTEND**	**- Tells learners to find out why earthworms are soft-bodied** **- Asks them to collect pictures of animals with different skeleton types** **- Prepares for next lesson on human skeleton**	**- Home assignment** **- Research**	**- Listens to home task** **- Prepares for next lesson**

CLASS WORK:

1. **Name the three types of skeletons found in animals.**
2. **What is an exoskeleton? Give two examples of animals with exoskeleton.**
3. **What is moulting? Why do animals moult?**
4. **Name two animals with hydrostatic skeleton.**
5. **Which type of skeleton do human beings have?**
6. **Complete the table:**

Type of Skeleton	Example Animal (1)	Example Animal (2)
Exoskeleton		

Hydrostatic		
Endoskeleton		

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 3

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Seven	**Science**		**80 Minutes**

THEME	**SUB-THEME**	**CONTENT**
The Human Body	**Muscular-Skeletal System**	**The human skeleton: main parts and functions**

COMPETENCES:

- **Subject Competence: The learner identifies the main parts of the human skeleton and explains their functions.**
- **Language Competence: The learner names bones and parts of the skeleton correctly using scientific terms.**

TEACHING METHODS/TECHNIQUES:

- **Use of charts/diagrams**
- **Demonstration**
- **Question and answer**
- **Labeling**
- **Discussion**
- **Note-taking**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Name the main parts of the human skeleton**
2. **State the functions of the human skeleton**
3. **Identify which parts protect which organs**

TEACHING/LEARNING MATERIALS:

- **Chart of human skeleton**

- **Model of human skeleton (if available)**
- **Flash cards with bone names**
- **Blank skeleton diagrams for labeling**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks: "What supports your body and gives it shape?" - Asks learners to feel their own bones (arm, ribs, skull) - Asks: "How many bones do you think are in the human body?" - Tells them: "Today we learn about the human skeleton"	- Question and answer - Self-discovery - Kinesthetic learning - Brainstorming	- Responds to greeting - Feels own bones - Answers questions - Guesses number of bones
20 mins	EXPLORE	- Displays chart of human skeleton - Divides class into groups - Gives each group a different part to focus on (skull, limbs, trunk, girdles) - Asks groups to observe and list the bones in their assigned part - Lets groups present their findings	- Group work - Observation - Guided discovery - Presentation	- Observes chart - Works in groups - Lists bones in assigned part - Presents findings to class
30 mins	EXPLAIN	- Points to each main part on the chart: - Skull: cranium, maxilla, mandible - Limbs: arms (humerus, radius, ulna), legs (femur, tibia, fibula) - Trunk: clavicle, sternum, ribs, vertebral column - Girdles: pelvis, scapula - Draws and labels each part on board - Explains functions of skeleton: - Protects delicate body organs - Gives shape to the body - Aids body movement - Gives support - Provides attachment for muscles - Bone marrow produces blood cells - Lists protective functions: - Ribcage protects heart and lungs - Skull protects brain - Pelvis protects bladder, uterus, ovaries - Eye socket protects eyes	- Lecture method - Demonstration - Use of charts - Drawing - Note-taking	- Listens attentively - Points to parts on own body - Copies notes - Asks questions - Copies diagrams

		- Backbone protects spinal cord - Mandible protects tongue		
15 mins	ELABORATE	- Gives blank diagram of human skeleton - Asks learners to label main parts - Asks: "Which organ is protected by the ribcage? By the skull?" - Provides matching exercise: match bone to its protective function - Discusses: "What would happen if we didn't have a skeleton?" - Guides learners to complete activities	- Labeling - Guided practice - Application - Discussion	- Labels diagram - Answers questions - Completes matching exercise - Participates in discussion - Writes in books
5 mins	EVALUATE	- Asks oral questions: - "Name four parts of the human skeleton" - "What does the skull protect?" - "What is the function of the ribcage?" - "Where are blood cells produced?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to find out more about bones - Asks them to memorize the names of major bones - Prepares for next lesson on types of bones	- Home assignment - Memorization task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What is the human skeleton?**
2. **Name four main parts of the human skeleton.**
3. **State three functions of the human skeleton.**
4. **Which part of the skeleton protects:**
   a) The brain?
   b) The heart and lungs?
   c) The spinal cord?
   d) The eyes?
5. **Draw and label the human skeleton showing at least 6 parts.**

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 4

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Bones in the human body: types of bones and their examples

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COMPETENCES:

- **Subject Competence: The learner identifies types of bones and gives examples of each.**
- **Language Competence: The learner uses scientific names for bones (e.g., femur, humerus, cranium).**

TEACHING METHODS/TECHNIQUES:

- **Use of charts/diagrams**
- **Demonstration**
- **Classification**
- **Question and answer**
- **Note-taking**
- **Labeling**
- **Game method**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Define a bone**
2. **Name the four types of bones**
3. **Give examples of each type of bone**
4. **Match bones to their other names**

TEACHING/LEARNING MATERIALS:

- **Chart showing types of bones**
- **Model skeleton (if available)**
- **Flash cards with bone names**
- **Pictures of different bone types**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks: "How many bones do you think are in the human body?" (adults 206, babies more) - Shows a long bone (picture or model) - Asks: "What type of bone is this? Where is it found?" - Introduces topic: types of bones	- Question and answer - Brainstorming - Use of pictures - Trivia	- Responds to greeting - Guesses number of bones - Observes bone picture - Answers questions - Listens
20 mins	EXPLORE	- Displays chart showing four types of bones - Divides class into four groups, each assigned one bone type - Asks groups to research (from chart) characteristics and examples of their bone type - Asks: "Where in the body are these bones found?" - Lets groups present their findings	- Group work - Observation - Guided discovery - Research - Presentation	- Observes chart - Works in assigned group - Identifies characteristics - Finds examples - Presents to class
30 mins	EXPLAIN	- Defines a bone: hard and rigid tissue found in vertebrates; formed by fusion of cartilages (ossification) - Explains four types of bones with diagrams: - Long bones: found in limbs; have yellow bone marrow; thick ends Examples: femur, humerus, radius, ulna, tibia, fibula - Short bones: found in feet and hands; have red bone marrow Examples: tarsals, carpals - Flat bones: flat shapes; protect organs Examples: cranium, scapula, pelvis, sternum - Irregular bones: complex shapes; found in back Examples: vertebral column (backbone), some facial bones - Explains ossification: process of bone formation	- Lecture method - Use of charts - Drawing - Question and answer - Note-taking	- Listens attentively - Takes notes - Names examples - Asks questions - Copies diagrams

		- Gives table of bones and their other names: Breastbone = sternum Shoulder blade = scapula Kneecap = patella Thigh bone = femur Hipbone = pelvis Skull = cranium Collarbone = clavicle Backbone = vertebral column/spine		
15 mins	ELABORATE	- Gives blank diagrams for learners to label different bone types - Asks learners to classify given bones into types - Plays matching game: match bone to its other name using flash cards - Discusses: "Why do babies have more bones than adults?" (bones fuse as they grow) - Asks learners to feel and identify bone types on their own bodies	- Labeling - Classification - Game method - Discussion - Kinesthetic learning	- Labels diagrams - Classifies bones - Participates in game - Answers questions - Feels bones on own body
5 mins	EVALUATE	- Asks oral questions: - "Name four types of bones" - "Give two examples of long bones" - "What is another name for the thigh bone?" - "What type of bone is the skull?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to memorize the other names of bones - Asks them to create flashcards for bone names at home - Prepares for diseases of bones	- Home assignment - Memorization task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What is a bone?**
2. **Name the four types of bones found in the human body.**
3. **Give two examples of:**
   a) Long bones
   b) Flat bones
   c) Short bones
   d) Irregular bones

4. **What is another name for:**
   a) Breastbone
   b) Shoulder blade
   c) Thigh bone
   d) Kneecap
   e) Collarbone
   f) Backbone
5. **Why do babies have more bones than adults?**

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 5

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Diseases and disorders of bones, health habits for bone health

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COMPETENCES:

- **Subject Competence: The learner identifies diseases and disorders affecting bones and explains how to keep bones healthy.**
- **Language Competence: The learner names diseases and describes symptoms and prevention.**

TEACHING METHODS/TECHNIQUES:

- **Use of charts/pictures**
- **Discussion**
- **Question and answer**
- **Storytelling**
- **Note-taking**
- **Case studies**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Name diseases that affect human bones**
2. **Name disorders that affect human bones**
3. **State health habits that keep bones healthy**

TEACHING/LEARNING MATERIALS:

- **Chart showing bone diseases and disorders**
- **Pictures of people with bone disorders (fractures, rickets, etc.)**
- **Calcium-rich food samples or pictures**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Tells short story about someone who broke a bone or has bone pain - Asks: "What can make our bones unhealthy?" - Shows picture of a person with bowed legs (rickets) - Asks: "What do you think caused this condition?" - Introduces topic: diseases and disorders of bones	- Storytelling - Question and answer - Picture study - Brainstorming	- Responds to greeting - Listens to story - Shares ideas - Observes picture - Answers questions
20 mins	EXPLORE	- Displays pictures of various bone diseases and disorders - Divides class into groups - Assigns each group a condition to research from the pictures/chart - Asks: "What are the signs? What might have caused it?" - Lets groups present their findings	- Picture study - Group work - Research - Presentation	- Observes pictures - Works in groups - Identifies signs - Suggests causes - Presents findings
30 mins	EXPLAIN	- Lists diseases that affect bones: - Polio: viral disease causing paralysis; can be prevented by immunization - Tuberculosis of the bones: bacterial infection affecting spine and joints - Leprosy: bacterial disease affecting skin, nerves, and bones - Bone cancer: malignant tumor in bones - Lists disorders that affect bones: - Fracture: break in bone (simple,	- Lecture method - Use of charts - Question and answer - Note-taking	- Listens attentively - Takes notes - Asks questions

		compound, greenstick) - Dislocation: bone displaced from joint - Osteoporosis: bones become weak and brittle (common in elderly) - Rickets: softening of bones in children due to vitamin D/calcium deficiency - Explains causes, symptoms, and prevention for each - Explains health habits for bone health: - Doing regular physical exercises - Feeding on foods rich in calcium and phosphorus (milk, fish, beans, leafy vegetables) - Having infants immunized against polio and tuberculosis - Observing correct body posture - Boiling drinking water to avoid polio infection - Getting enough vitamin D from sunlight		
15 mins	ELABORATE	- Asks learners to name calcium-rich foods from their locality - Discusses: "Why is immunization important for bone health?" - Discusses: "How can we prevent fractures at home and school?" - Shows first aid for fractures (splints, immobilization) - Asks learners to demonstrate how to care for a fracture victim - Guides learners to write a summary of bone health tips	- Discussion - Question and answer - Demonstration - Application	- Names calcium-rich foods - Participates in discussion - Suggests prevention methods - Observes demonstration - Practices first aid - Writes summary
5 mins	EVALUATE	- Asks oral questions: - "Name two diseases that affect bones" - "What is a fracture? What are the types?" - "What is rickets? What causes it?" - "Give three ways of keeping bones healthy" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to find out about joints in the body - Asks them to list calcium-rich foods they eat at home - Prepares for next lesson on joints	- Home assignment - Research	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **Name two diseases that affect human bones.**
2. **What is a fracture? Distinguish between simple and compound fracture.**
3. **What is dislocation?**
4. **What is rickets? What causes it?**
5. **Give three health habits that keep bones healthy.**
6. **Why is calcium important for bones? Name three calcium-rich foods.**

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 6

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Joints in the human body: definition, importance, classification

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COMPETENCES:

- **Subject Competence: The learner defines joints, states their importance, and classifies them as movable or immovable.**
- **Language Competence: The learner uses terms like joint, movable, immovable, hinge, ball and socket correctly.**

TEACHING METHODS/TECHNIQUES:

- **Demonstration**
- **Use of charts**
- **Question and answer**
- **Kinesthetic learning**
- **Note-taking**
- **Modeling**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Define a joint**

2. **State the importance of joints**
3. **Classify joints as movable or immovable**

TEACHING/LEARNING MATERIALS:

- **Chart showing joints in human body**
- **Model skeleton (if available)**
- **Door hinge (for demonstration)**
- **Ball and socket model (or picture)**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks learners to bend their arms, nod their heads, move their legs - Asks: "What allows these parts to move?" - Shows a door hinge and demonstrates its movement - Asks: "How is this similar to parts of your body?" - Introduces topic: joints	- Kinesthetic learning - Question and answer - Demonstration - Analogy	- Responds to greeting - Moves body parts - Answers questions - Observes door hinge - Makes connections
20 mins	EXPLORE	- Asks learners to work in pairs and identify all the places where bones meet in their bodies - Asks them to move each joint and describe the type of movement - Provides chart for reference - Asks: "Which joints move a lot? Which move a little? Which don't move at all?" - Lets pairs share findings	- Pair work - Discovery method - Observation - Kinesthetic learning	- Works in pairs - Identifies joints on own body - Describes movements - Shares findings
30 mins	EXPLAIN	- Defines a joint: body part where two or more bones meet - Draws diagram showing different joints - Explains importance of joints: - Allow body movement - Provide flexibility to the human body - Enable us to perform various activities - Classifies joints into: - Movable joints: allow movement	- Lecture method - Demonstration - Use of charts - Drawing - Note-taking	- Listens attentively - Takes notes - Copies diagrams - Asks questions

		(found in limbs, neck, etc.) - Immovable joints: do not allow movement; also called fibrous joints - Gives example of immovable joints: suture joints of the skull - Shows diagrams of both types - Explains that joints have structures (ligaments, cartilage, synovial fluid) that help them function smoothly		
15 mins	ELABORATE	- Asks learners to identify movable and immovable joints on their bodies and on diagrams - Asks: "Where in the body do we find immovable joints? Why are they immovable?" - Discusses why skull bones are fixed (to protect the brain) - Asks: "What would happen if our skull had movable joints?" - Guides learners to create a table comparing movable and immovable joints	- Application - Question and answer - Discussion - Comparison	- Identifies joints on own body and diagrams - Answers questions - Participates in discussion - Creates comparison table
5 mins	EVALUATE	- Asks oral questions: - "What is a joint?" - "Why are joints important?" - "What is the difference between movable and immovable joints?" - "Where are immovable joints found?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to find out types of movable joints - Asks them to note which joints in their body move in different ways - Prepares for next lesson on types of movable joints	- Home assignment - Observation task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What is a joint?**
2. **Give two importance of joints in the human body.**
3. **What are movable joints?**
4. **What are immovable joints? Give one example.**
5. **Where in the body are immovable joints found?**
6. **Complete the table:**

Type of Joint	Characteristics	Example in Body
Movable joints		
Immovable joints		

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 7

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Types of movable joints: Hinge joints, Ball and socket joints

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COMPETENCES:

- **Subject Competence: The learner identifies hinge joints and ball and socket joints and gives examples of where they are found.**
- **Language Competence: The learner describes the movement of each type of joint.**

TEACHING METHODS/TECHNIQUES:

- **Demonstration**
- **Use of charts**
- **Question and answer**
- **Kinesthetic learning**
- **Note-taking**
- **Modeling**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Describe hinge joints and give examples**
2. **Describe ball and socket joints and give examples**
3. **Explain functions of parts of a joint**

TEACHING/LEARNING MATERIALS:

- **Chart showing hinge and ball and socket joints**
- **Model skeleton (if available)**
- **Door hinge (real or picture)**
- **Ball and socket model (or picture)**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Shows a door hinge and demonstrates how it moves - Asks: "How does this move? In how many directions?" - Asks learners to move their elbows and knees - Asks: "How do these move? Are they like the door hinge?" - Shows a ball and socket model (or a shoulder joint model) - Asks: "How is this different from the hinge?" - Introduces topic: types of movable joints	- Realia - Demonstration - Question and answer - Comparison	- Responds to greeting - Observes door hinge - Moves elbows and knees - Answers questions - Observes ball and socket model - Identifies differences
20 mins	EXPLORE	- Divides class into groups - Asks groups to explore movement of different joints on their bodies: - Elbow and knee (hinge) - Shoulder and hip (ball and socket) - Asks them to record: - Direction of movement - Range of motion - Examples of activities using each joint - Lets groups present findings	- Group work - Kinesthetic learning - Discovery method - Observation - Presentation	- Works in groups - Explores joint movements - Records observations - Presents findings
30 mins	EXPLAIN	- Explains hinge joints: - Move in only one plane (like a door hinge) - Allow bending and straightening only - Found in knees, elbows, knuckles, ankles - Examples: elbow joint, knee	- Lecture method - Demonstration - Use of charts - Drawing - Note-taking	- Listens attentively - Takes notes - Copies diagrams - Asks questions

		joint, finger joints - Draws diagram of hinge joint - Explains ball and socket joints: - Move in more than one plane (three planes: forward, backward, sideways, rotation) - Allow all-round movement - Found at shoulders and hips - Examples: shoulder joint, hip joint - Draws diagram of ball and socket joint - Explains functions of joint parts: - Synovial fluid: reduces friction by lubricating the joint - Ligament: provides attachment for bones at a joint; holds bones together - Tendon: provides attachment for bone and muscle - Cartilage: reduces friction by preventing bones from rubbing directly; acts as shock absorber - Joint capsule: encloses the joint and contains synovial fluid		
15 mins	ELABORATE	- Asks learners to identify hinge and ball and socket joints on their bodies and on diagrams - Asks: "What would happen if our knee was a ball and socket joint?" (unstable, would bend sideways) - Discusses importance of synovial fluid and cartilage in reducing friction - Asks learners to demonstrate activities that use each type of joint - Provides worksheet with diagrams to label and identify joint types	- Application - Question and answer - Discussion - Kinesthetic learning - Worksheet	- Identifies joints on body and diagrams - Answers questions - Participates in discussion - Demonstrates activities - Completes worksheet
5 mins	EVALUATE	- Asks oral questions: - "What is a hinge joint? Give two examples" - "What is a ball and socket joint? Give two examples" - "What is the function of	- Oral assessment - Written assignment	- Answers oral questions - Copies class work

		cartilage at a joint?" - "What does synovial fluid do?" - Gives class work		
5 mins	EXTEND	- Tells learners to find out about gliding and pivot joints - Asks them to note movements in their wrists and neck - Prepares for next lesson	- Home assignment - Observation task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What is a hinge joint? Give two examples.**
2. **What is a ball and socket joint? Give two examples.**
3. **State the function of:**
   a) Synovial fluid
   b) Ligament
   c) Cartilage
   d) Tendon
4. **Where in the body are ball and socket joints found?**
5. **Complete the table:**

Type of Joint	Movement Allowed	Examples (2 each)
Hinge joint		
Ball and socket		

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 8

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Types of movable joints: Gliding joints, Pivot joints, Immovable joints

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COMPETENCES:

- **Subject Competence: The learner identifies gliding joints, pivot joints, and immovable joints and gives examples.**

- **Language Competence: The learner describes the movement of each type of joint.**

TEACHING METHODS/TECHNIQUES:

- **Demonstration**
- **Use of charts**
- **Question and answer**
- **Kinesthetic learning**
- **Note-taking**
- **Modeling**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Describe gliding joints and give examples**
2. **Describe pivot joints and give examples**
3. **Describe immovable joints and give examples**
4. **Identify joints in diagrams**

TEACHING/LEARNING MATERIALS:

- **Chart showing gliding, pivot, and immovable joints**
- **Model skeleton (if available)**
- **Skull model or picture**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks learners to move their wrists and ankles in different directions - Asks: "How do these joints move? Do they move like hinges?" - Asks learners to turn their heads side to side, nod up and down - Asks: "What allows this movement?" - Asks: "Can you move your skull bones? Why not?" - Introduces topic: gliding, pivot, and immovable joints	- Kinesthetic learning - Question and answer - Demonstration	- Responds to greeting - Moves wrists and ankles - Turns head - Answers questions - Attempts to move skull

20 mins	**EXPLORE**	**- Divides class into three groups** **- Assigns each group a joint type to investigate:** **Group 1: Gliding joints (wrists, ankles)** **Group 2: Pivot joints (neck)** **Group 3: Immovable joints (skull)** **- Asks groups to explore movement and record characteristics** **- Lets groups present findings with demonstrations**	**- Group work** **- Kinesthetic learning** **- Discovery method** **- Observation** **- Presentation**	**- Works in assigned group** **- Explores joint movement** **- Records characteristics** **- Demonstrates to class** **- Presents findings**
30 mins	**EXPLAIN**	**- Explains gliding joints:** **- Allow flat bones to slide over one another** **- Also called plane/sliding joints** **- Allow movement in two planes (back-forth, side-to-side)** **- Found at wrist (between carpals) and ankle (between tarsals)** **- Also between vertebrae** **- Draws diagram of gliding joint** **- Explains pivot joints:** **- Allow one bone to rotate on another bone** **- Also called rotary joints** **- Bones involved: atlas (first vertebra) and axis (second vertebra)** **- Allow head to nod, rotate, turn, shake freely** **- Found at neck (between atlas and axis)** **- Draws diagram of pivot joint** **- Reviews immovable joints:** **- Do not allow movement** **- Also called fibrous joints** **- Bones are fixed together by fibrous tissue** **- Found in skull (suture joints)** **- Also called sutures** **- Shows diagrams of each** **- Summarizes all joint types in a table**	**- Lecture method** **- Demonstration** **- Use of charts** **- Drawing** **- Note-taking**	**- Listens attentively** **- Takes notes** **- Copies diagrams** **- Asks questions**
15 mins	**ELABORATE**	**- Asks learners to identify gliding, pivot, and immovable joints on diagrams and on their bodies** **- Asks: "What would happen if our neck had a hinge joint?" (couldn't rotate head)** **- Discusses importance of each joint type for daily activities**	**- Application** **- Question and answer** **- Discussion** **- Worksheet**	**- Identifies joints on diagrams and body** **- Answers questions** **- Participates in discussion** **- Completes worksheet**

		- Provides worksheet with diagrams to label and identify joint types - Asks learners to complete a comparison table of all joint types		- Creates comparison table
5 mins	EVALUATE	- Asks oral questions: - "What is a gliding joint? Give two examples" - "What is a pivot joint? Where is it found?" - "Where are immovable joints found?" - "What bones make up the pivot joint at the neck?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to review all types of joints for a quiz - Asks them to create a poster showing different joint types - Prepares for next lesson on muscles	- Home assignment - Creative task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What is a gliding joint? Give two places where it is found.**
2. **What is a pivot joint? Where is it found?**
3. **What are immovable joints? Give one example.**
4. **Name the two bones that make up the pivot joint at the neck.**
5. **Complete the table of all joint types:**

Joint Type	Movement Allowed	Examples (2 each)
Hinge joint		
Ball and socket		
Gliding joint		
Pivot joint		
Immovable joint		

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 9

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Muscles in the human body: types of muscles, voluntary muscles, antagonistic muscles

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COMPETENCES:

- **Subject Competence: The learner identifies types of muscles and explains how antagonistic muscles work.**
- **Language Competence: The learner uses terms like voluntary, involuntary, biceps, triceps, contract, relax correctly.**

TEACHING METHODS/TECHNIQUES:

- **Demonstration**
- **Use of charts**
- **Question and answer**
- **Model demonstration**
- **Note-taking**
- **Kinesthetic learning**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Define muscles**
2. **Distinguish between voluntary and involuntary muscles**
3. **Explain how antagonistic muscles work**
4. **Give examples of voluntary muscles**

TEACHING/LEARNING MATERIALS:

- **Chart showing muscles of the body**
- **Model of arm showing biceps and triceps**
- **Rubber band (to demonstrate contraction)**
- **String and rubber tube model**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks learners to bend and straighten their arms repeatedly - Asks: "What do you feel happening in your arm?" (muscles bulge and relax) - Shows a rubber band being stretched and released - Relates to muscles contracting and relaxing - Asks: "Can you control all your muscles? Can you stop your heart?" - Introduces topic: muscles	- Kinesthetic learning - Demonstration - Question and answer - Analogy	- Responds to greeting - Bends and straightens arm - Feels muscles - Observes rubber band - Answers questions - Listens
20 mins	EXPLORE	- Divides class into pairs - Asks learners to take turns: - Feeling partner's biceps when arm is bent and straight - Feeling calf muscles when standing on toes - Feeling jaw muscles when clenching teeth - Asks: "What happens to the muscle when you bend? When you straighten?" - Asks: "Which muscles can you control? Which work automatically?" - Lets pairs share observations	- Pair work - Kinesthetic learning - Discovery method - Observation	- Works in pairs - Feels muscles in different positions - Observes changes - Discusses observations - Shares with class
30 mins	EXPLAIN	- Defines muscles: fibrous elastic tissues that contract and relax to produce movement; attached to bones by tendons - Draws diagram of muscle structure - Explains voluntary muscles: movement controlled by one's will - Examples: biceps, triceps, hamstring, quadriceps, calf muscles, pectoral muscles, gluteus muscles, abdominal muscles - Also called skeletal muscles - Explains antagonistic muscles: muscles that work in pairs but have opposite effects - Example: biceps (flexor) and	- Lecture method - Demonstration - Use of charts - Drawing - Note-taking	- Listens attentively - Takes notes - Copies diagrams - Asks questions

		triceps (extensor) in the arm - Demonstrates action with model or diagram: - When arm is bent/raised: biceps contract (shorten), triceps relax (lengthen) - When arm is straightened: biceps relax, triceps contract - Draws diagram showing biceps and triceps action - Explains other antagonistic pairs: quadriceps and hamstrings (thigh), pectorals and trapezius (shoulder)		
15 mins	ELABORATE	- Shows diagram of arm muscles - Asks learners to explain what happens during bending and straightening - Asks: "Why do muscles work in pairs?" (can only pull, not push) - Provides model with string and rubber tube to demonstrate antagonistic action - Asks learners to create a simple model at their desks - Discusses importance of antagonistic muscles for smooth movement - Asks learners to identify other antagonistic muscle pairs in the body	- Application - Question and answer - Modeling - Discussion	- Explains muscle action - Answers questions - Creates simple model - Participates in discussion - Identifies other muscle pairs
5 mins	EVALUATE	- Asks oral questions: - "What are muscles?" - "What are voluntary muscles? Give two examples" - "What are antagonistic muscles?" - "What happens to biceps and triceps when you bend your arm?" - "What happens when you straighten your arm?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to find out about involuntary muscles - Asks them to notice which muscles work without their control - Prepares for next lesson	- Home assignment - Observation task	- Listens to home task - Prepares for next lesson

CLASS WORK:

1. **What are muscles?**
2. **What are voluntary muscles? Give three examples.**
3. **What are antagonistic muscles? Give one example.**
4. **Describe what happens to the biceps and triceps when:**
   a) The arm is bent
   b) The arm is straightened
5. **Why do muscles work in pairs?**
6. **Draw and label a diagram showing the biceps and triceps in the arm.**

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 10

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Science		80 Minutes

THEME	SUB-THEME	CONTENT
The Human Body	Muscular-Skeletal System	Involuntary muscles and importance of muscles

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COMPETENCES:

- **Subject Competence: The learner identifies involuntary muscles and explains the importance of muscles in the body.**
- **Language Competence: The learner uses terms like cardiac, intercostal, diaphragm, peristalsis correctly.**

TEACHING METHODS/TECHNIQUES:

- **Use of charts**
- **Question and answer**
- **Discussion**
- **Note-taking**
- **Demonstration**

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. **Define involuntary muscles**
2. **Give examples of involuntary muscles**
3. **State the importance of muscles in the human body**

TEACHING/LEARNING MATERIALS:

- **Chart showing involuntary muscles**
- **Diagram of heart, digestive system, lungs**
- **Stethoscope (if available)**
- **Chalkboard/whiteboard**

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	ENGAGE	- Greets learners - Asks learners to place hand on chest and feel heartbeat - Asks: "Can you make your heart stop beating by will? Can you make it beat faster?" - Asks: "Can you control your stomach churning? Your breathing completely?" - Asks: "What type of muscles do you think controls these?" - Introduces involuntary muscles	- Question and answer - Self-observation - Brainstorming	- Responds to greeting - Feels heartbeat - Answers questions - Thinks about body functions - Listens
20 mins	EXPLORE	- Divides class into groups - Assigns each group a body function to research from charts: Group 1: Heartbeat Group 2: Breathing Group 3: Digestion Group 4: Blinking - Asks: "What muscles are involved? Can you control them?" - Lets groups present findings	- Group work - Research - Discussion - Presentation	- Works in groups - Researches assigned function - Identifies muscles involved - Presents findings
30 mins	EXPLAIN	- Defines involuntary muscles: muscles whose movement is not controlled by one's will; they work automatically - Draws diagram showing locations of involuntary muscles	- Lecture method - Use of charts - Question and answer - Drawing - Note-taking	- Listens attentively - Takes notes - Copies diagrams - Asks questions

		- Gives examples with functions: - Cardiac muscles: found only in heart; contract rhythmically to pump blood; never tire - Intercostal muscles: between ribs; aid breathing by lifting and lowering rib cage - Diaphragm muscles: sheet of muscle below lungs; contracts and relaxes to aid breathing - Intestinal muscles: in walls of intestines; produce peristalsis to move food along - Sphincter muscles: circular muscles that open and close body openings (anus, bladder) - Eyelid muscles: control blinking (both voluntary and involuntary) - Explains importance of muscles: - Aid body movement by contracting and relaxing - Store oxygen with help of myoglobin (in muscle tissue) - Intercostal muscles and diaphragm aid breathing - Muscles of alimentary canal aid peristalsis (wave-like movements) - Heart muscles facilitate circulation of blood throughout body - Maintain posture (back and neck muscles) - Produce heat (shivering when cold)		
15 mins	ELABORATE	- Asks learners to name activities controlled by involuntary muscles - Discusses: "What would happen if our heart muscles were voluntary?" (we'd forget to beat, die) - Discusses: "Why do we sometimes blink voluntarily and sometimes involuntarily?" - Demonstrates peristalsis using a tube and ball - Asks learners to feel their pulse and breathing rate before and after exercise - Discusses how involuntary muscles respond to body needs	- Discussion - Question and answer - Demonstration - Application	- Names activities - Participates in discussion - Answers questions - Observes demonstration - Measures pulse and breathing - Records observations

5 mins	EVALUATE	- Asks oral questions: - "What are involuntary muscles?" - "Give three examples of involuntary muscles and where they are found" - "What is peristalsis?" - "State two importance of muscles in the body" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	EXTEND	- Tells learners to review all topics on muscular-skeletal system - Asks them to create a mind map summarizing the entire topic - Prepares for next topic: Disorders of muscles	- Home assignment - Creative task	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. **What are involuntary muscles?**
2. **Give three examples of involuntary muscles and state where each is found.**
3. **What is peristalsis? Which muscles are responsible for it?**
4. **State three importance of muscles in the human body.**
5. **Complete the table:**

Type of Muscle	Voluntary or Involuntary?	Location	Function
Biceps			
Cardiac			
Intercostal			
Triceps			
Intestinal			
Diaphragm			

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

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TO PURCHASE COMPLETE AND MORE FROM P.1-P.7

SING: Mr. Hyson